WASTE (INHERITANCE) #REPORT5 BY PETROS TATSIOPOULOS

As a sculpture assistant for the project, artist Petros Tatsiopoulos has been working with Nelson's aluminum and metallic waste. This post describes his research, discoveries and personal thoughts about the material after working intensively with it for the last four months.

Photo by Petros Tatsiopoulos

Working with aluminum and other metals that exist in product waste packaging seemed like a straight-forward objective. That is because most aluminum products are in a single recycling category and the other metals included are only iron and its alloys, like steel and tinned final products. However, I discovered that it is not only the alloys that can make working with metallic waste complicated but also its combination with plastic or paper. Products we overlook include Tetrapack packaging or bags for coffee, potato chips and other. I worked with each variant from a practical point of view, taking into consideration brittleness, elasticity, ways of combination, bending or sewing. My sculptural objectives have been to create combinations of waste to form fabrics, armor-like components, or to be used structurally. I've also needed to deconstruct the elements of different packaging, not only to reduce their volume but also to find and shape new components in order to build, stack on, or store in the sculpture.

Photo by Petros Tatsiopolos. Materials broken down and made compact.

Photo by Petros Tatsiopoulos. Iron can bottoms made into fabric by Petros Tatsiopoulos.

I've experienced working with all this metal physically and psychologically. To start with, aluminum and tinned metal packaging products are very shiny in their interior and sometimes on their exterior. This has had an immediate and prolonged effect on my somatic and psychological state while working with these materials. It has felt sometimes as if I were living in a shiny world of sharp metal, where the effects of the reflective surface will never go away–the feeling of working in a gleaming, endless loop. The sheer number of cans found in a small family's waste made me have negative feelings about the individuation of portions. Not only because I had to work repetitively with the cans, but also in a wider sense for the amount of waste we produce using so much packaging. Metal, going from aluminum to steel in brittleness and hardness, has taken a toll on my body–especially the hands and back– having to cut and deconstruct it. The deconstruction of packaging has required that I find and use differentiated methods to process different products.

Photo by Petros Tatsiopoulos. Aluminum can fabric by Petros Tatsiopoulos

On a systemic level, recycling regulations in Europe have increased since the start of the millennium. Recycling consensus grows and the industrial world has steadily shifted towards including sustainability issues in its agenda. However, according to the Global Recycling magazine website, Greece lags behind. In 2018, recycling in Greece accounted for just 13,2% of trash, while 83,6% of trash sent to landfills was recyclable waste. In the same year, aluminum can recycling in Greece was calculated at 55%. Given the environmental cost of new aluminum, however, recycling is essential.

Bauxite. https://www.mdpi.com/2077-1312/9/11/1289/htm

Aluminum, although not a rare metal, is only found in nature as an ore that must be extracted and refined. Bauxite typically has the highest concentration of aluminum–the reason why it is mined. Filtering aluminum oxide from bauxite is a process that creates an alkaline residue referred to as red mud. This is often stored indefinitely in landfills or dams, which can fail. Plans to manage it include its conversion to building materials or the reclamation of heavy metals. Finally, the separation of aluminum from oxygen is a process that requires vast amounts of electrical energy. When not stored properly, red mud is a significant hazard for the environment. As seen in Bert Ehgartner's documentary about the aluminum industry, bauxite mining in the Amazon and elsewhere, results in deforestation; the dumping of caustic red sludge; the poisoning of water which has a toll on the life of indigenous and other low-income communities; and long term health problems for workers in the facility–some later, others sooner.

Κόκκινη λάσπη. https://www.mdpi.com/2077-1312/9/11/1289/htm

Although there are some preventative or rehabilitating measures, like reforestation projects or the filtration of refinement waste from its caustic chemicals, in my view, these problems are treated superficially if they are addressed at all. The intake of aluminum has been proven to be neurotoxic for the brain and body causing various diseases like dementia, muscle-related problems and other.[1] According to Ehgartner's documentary, scientists and physicians working in the area of aluminum research are not convinced that the aluminum intake present in the consumer world, through vaccines, medication, deodorants and water-filtration, is safe and are not contributing to the health problems mentioned above.

Through my research I've learned that Greece was and continues to be one of the main mining places for bauxite in Europe. Lena Papastefanaki writes about the role bauxite mining played in Greece during the second World War, which was also a war of resources. The German army occupied Greece and its bauxite resources, like the mining areas around Delphi, to ship the resource back to Germany.[2] History continues, only now corporations vie to control resources. This part of industrial history, if I may call it so, is only one of a series of industrial revolutions beginning around 1770 and coming to age with the present, according to Mastrogeorgiou, fourth industrial revolution.[3] This fourth development involves the extended use of robots, machine learning and artificial intelligence and the corresponding ever-expanding need for bauxite-based products. Through this timeline, it is evident to me that industrial growth has been ruthless and rapid since its beginning, associated with a totalitarian problem-solving type of mentality, in a profit-driven frenzy, with only secondary consideration for environmental or health problems that surface or might surface through its actions.

Photo by Anna Primou. Armor from beer can tops by Petros Tatsiopoulos

For certain, working on this project has changed the way I handle my own recyclable waste. I've started paying more attention to cleaning, separation, and identification of the waste I produce. Also and through that, the plan to avoid or at least minimize non-recyclable packaging has started to appear in my mind. Knowing the nature of this waste is not only important in terms of sending it to a potential recycler but also in starting to use the packaging waste a second or third time, or in the making of something at home and upcycling these materials at the individual level. But on a more systemic level, I find an analogy for sustainability and the way the industrial world behaves in the Anthropocene era in the tale of The Emperor's New Clothes. Like the emperor who doesn't realize his flawed premise and vulnerability, the industrial world, e.g. aluminum industry, produces with profit as a primary gear, paying secondary (or no) attention to the real problems of the industry. To change this would require an awareness and understanding of overproduction, repression of local communities, environmental limits and vulnerabilities, and a reflection on what growth means in terms of sustainability for all. Seeing this clearly and holistically could give rise to an evolved understanding of what industry could mean.

Petros Tatsiopoulos studied Architecture (KIT Karlsruhe, Germany) and Scenography and Exhibition Design (HfG Karlsruhe, Germany) and has a BA in Visual Arts (ACG Athens, Greece). He has participated in various art projects in theater scenography (Baden State Theater), artist run spaces, group exhibitions and festivals (InSonic 2015). He has been preoccupied with sound art and sculpture since 2014 and the main focus of his art has been mathematical abstraction and a philosophical exploration on the nature of art and its place in social structures.

[1] Jelenković, Ankica. *Aluminum Neurotoxicity : From Subtle Molecular Lesions to Neurological Diseases*. Neurology–Laboratory and Clinical Research Development Series. New York: Nova Science Publishers, Inc, 2016.

[2] Lena Papastefanaki, "Greece Has Been Endowed by Nature with This Precious Material" in Aluminum Ore : The Political Economy of the Global Bauxite Industry, ed. Gendron, Robin S., Ingulstad, Mats, and Storli, Espen. Vancouver: UBC Press, 2013.

[3] Μαστρογεωργίου, Γιάννης [Mastrogeorgiou, Giannis]. Τέταρτη βιομηχανική επανάσταση [Fourth Industrial Revolution]. Athens: Φιλελεύθερος Τύπος Α.Ε., 2019, ISBN: 978-618-84245-7-9.

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